

Code Refactors

1) Two classes, user and player, were too similar, had the same fields, had mostly the same methods and covered the same things.

The screenshot shows a GitHub issue interface. At the top, it says 'aryagith opened last week · edited by sleimon'. The issue title is 'Code Smell ID: 1' with a category of 'code smell'. The description explains that the `User` and `Player` classes are redundant, violating the DRY principle. A refactoring suggestion is provided to merge them into a single class. The severity is marked as 'High'. The issue has a history of updates: 'aryagith' added the label 'code smell', 'sleimon' assigned it to himself and 'aryagith', 'sleimon' added three commits (498df12, 494bb12, 6b132e0) that reference the issue, and 'sleimon' mentioned it 9 hours ago with a link to a commit. A comment by 'sleimon' 9 hours ago states that a new feature was added to split the functionalities of both classes. Finally, 'sleimon' closed the issue as 'completed' 9 hours ago.

aryagith opened last week · edited by sleimon

- **Code Smell ID:** 1
- **User Story:** User Profile and Chips/Virtual Currency Management
- **Description:**
The `User` and `Player` classes are redundant and contain overlapping functionality and data. Both classes store similar data (e.g., chip count, wins, losses), which leads to redundancy and unnecessary complexity. The duplication of this data violates the **DRY (Don't Repeat Yourself) principle** and creates unnecessary coupling between the two classes. This design also leads to potential issues with data synchronization and consistency between the two classes. It also makes the code in the game class much less readable and has contributed to some of the bugs in the game class.
- **Refactoring Suggestion:**
Merge the `User` and `Player` classes into a single class. This will remove redundancy and improve code maintainability by consolidating the related data into one place. The class should manage the user-specific data (e.g., chips, wins, losses) while keeping logic related to gameplay (e.g., betting, win/loss outcomes) in the same class.
- **Severity:** 🚨 High

Create sub-issue

aryagith added **code smell** last week

sleimon assigned [sleimon](#) and [aryagith](#) last week

sleimon added 3 commits that reference this issue 5 days ago

Laid the foundation for the player class to absorb the user class and...	498df12
Laid the foundation for the player class to absorb the user class and...	494bb12
Added a new feature, lifetime wins/losses and session wins/losses	6b132e0

sleimon mentioned this 9 hours ago

🔗 [Added a new feature, lifetime wins/losses and session wins/losses #68](#)

sleimon 9 hours ago

Added a new feature that adds a different identity to the player class
Now the player class keeps track of session wins/losses
And the user class keeps track of lifetime wins and losses
Which makes each class non-redundant

sleimon closed this as **completed** 9 hours ago

To solve this problem, we split the functionalities of both classes such that they had their own identities and nothing was overlapping.

2) Our game class was over 700 lines long and handled both business logic and GUI functionalities

Code Smell: Large Class #31

Closed

aryagith opened last week · edited by sleimon

Code Smell ID: 3

- **User Story:** Different Cards (Deck, Hand, and Game Functionality) and Betting
- **Description:**
The `Game.java` class is too large and handles too many responsibilities, including **GUI, game logic, and statistics management**. This violates the **Single Responsibility Principle (SRP)** and makes the code hard to maintain.
- **Refactoring Suggestion:**
Break `Game.java` into separate classes:
 - `GameGUI` for GUI-related logic
 - `GameLogic` for game rules and mechanics
 - `GameStats` for statistics tracking
- **Severity:** 🚨 High

Create sub-issue

sleimon assigned sleimon and aryagith last week

aryagith added code smell last week

sleimon mentioned this 2 days ago

Code Smell: Long Methods: UI, Authentication, and Game Logic in Main Class #41

sleimon removed their assignment 1 hour ago

sleimon 1 hour ago

Owner

Closed by Arya in his most recent pushes to his branch (and eventually the main branch).
He split the game class into two classes (`gameLogic` and `gameGUI`) which solved the large class issue

sleimon closed this as completed 1 hour ago

We solved this by splitting the game class into two classes, `gameLogic` and `gameGUI`, one handles the business logic and the other handles the GUI.

3) A method in the game class was around 300 lines and covered multiple functions such as creating buttons, assigning JLabels and setting up event listeners

Code Smell: Large Method (setupGUI in Game.java) #32

Closed

aryagith opened last week · edited by sleimon

Code Smell ID: 4

- **User Story:** Different Cards (Deck, Hand, and Game Functionality) and Chips/Virtual Currency Management
- **Description:**
The `setupGUI()` method in `Game.java` is too long and handles too many responsibilities, including button creation, event handling, and game statistics display.
- **Refactoring Suggestion:**
Split `setupGUI()` into smaller methods, each with a single responsibility, such as:
 - `createButtons()`
 - `addEventListeners()`
 - `displayGameStats()`
- **Severity:** ⚠ Low

Create sub-issue

aryagith added `code smell` last week

sleimon assigned `sleimon` and `aryagith` last week

sleimon added 2 commits that reference this issue 2 days ago

Fixed some ticket #32, code smell ID #4

Merge pull request #58 from sleimon/sleiITR3

cc56bb2

Verified bea1de5

sleimon 2 days ago

Owner

Split the `setupGUI()` method into three methods.

- `setupGUI()` handles the size of the `JPanel`, displaying the card pngs and the creation and placement of all the `JLabels`
- `createButtons()` handles the creation of the many buttons as well as the size and placement of them
- Lastly, `actionListeners` handle all the event listeners on all the buttons

We solved this problem by splitting the `setupGUI` method into three

`setupGUI` handles the size of the `JPanel`, the card pngs and the creation and placement of all the `JLabels`

`createButtons()` handles the creation of the many buttons as well as the size and placement of them

Lastly, `actionListeners` handle all the event listeners on all the buttons

4)

There were too many primitive values in the game class.

Code Smell: Primitive Obsession #30

Closed



aryagith opened last week · edited by aryagith

Edits ...

Code Smell ID: 2

- **User Story:** User Profile
- **Description:**
The `Game.java` class uses primitive fields like `pushes`, `roundsPlayed`, and `startingChips` for tracking game state. These could be better represented with meaningful objects instead of primitive types.
- **Refactoring Suggestion:**
Replace these primitive variables with a `GameState` class that encapsulates all relevant game state data. Use abstractions for `deck`, `discarded`, `dealer`, and `player` for better management.
- **Severity:** ⚠ Medium

Create sub-issue



aryagith added `code smell` last week



sleimon assigned [sleimon](#) and [aryagith](#) last week



sleimon added a commit that references this issue 2 days ago

Fixed some code smells ...

d540aa2



sleimon mentioned this 2 days ago

Fixed some code smells #57



sleimon 2 days ago

Owner ...

Those primitive fields were dead code anyways, I just removed them.



We solved this by deleting them since we refactored the class to not need all those primitive values.

5) Lots of unused and commented out code in the game class

Code Smell: Dead Code #33

Closed

aryagith opened last week · edited by sleimon

Code Smell ID: 5

- **User Story:** User Profile , Betting, Chips/Virtual Currency Management
- **Description:**
`Game.java` contains large segments of commented-out code that are no longer in use, cluttering the codebase and reducing readability.
- **Refactoring Suggestion:**
Remove all unused and commented-out code to enhance clarity and maintainability.
- **Severity:** ⚠ Low

Create sub-issue

aryagith added `code smell` last week

sleimon assigned [sleimon](#) and [aryagith](#) last week

sleimon added a commit that references this issue 2 days ago
Fixed some code smells `d540aa2`

sleimon mentioned this 2 days ago
[Fixed some code smells #57](#)

sleimon 2 days ago

Removed the unused code and code that have been commented out.

Solved this by removing the dead and uncommented code

6) There was some duplicate code regarding the wins, losses, and pushes fields in the game class

Code Smell: Duplicate Code #34

Closed

aryagith opened last week · edited by sleimon

Code Smell ID: 6

- **User Story:** User Profile , Betting, Statistics
- **Description:**
The logic for updating user statistics (wins , losses , pushes) is repeated across multiple methods in `Game.java` , violating the DRY principle.
- **Refactoring Suggestion:**
Create a single method to handle all stat updates and call it wherever needed.
- **Severity:** ⚠ Medium

Create sub-issue

aryagith added code smell last week

sleimon assigned sleimon and aryagith last week

sleimon added a commit that references this issue 2 days ago

Fixed some code smells d540aa2

sleimon mentioned this 2 days ago

Fixed some code smells #57

sleimon 2 days ago

Owner

I created a method in the player class that'll update wins, losses or pushes by 1. When we successfully dismantle the user class and reinstate the player class in the game class as the class that tracks certain things for whomever is playing, this will be resolved.

We solved this by creating a method that'll do the operation needed on the wins, losses or pushes fields without calling those fields excessively.

7) The game class is overly dependent on the user class since the game class frequently accesses the user class

Code Smell: Feature Envy (Game.java Accessing User Data Directly)

Closed

aryagith opened last week

Code Smell ID: 7

- **User Story:** User Profile
- **Description:**
The `Game.java` class frequently accesses `User` class properties (`wins`, `losses`, `chips`), making `Game` overly dependent on `User`.
- **Refactoring Suggestion:**
Move logic for updating/retrieving user stats (`wins`, `losses`, `chips`) to `User` and let `Game` interact through method calls instead.
- **Severity:** Medium

Create sub-issue

aryagith added `code smell` last week

sleimon self-assigned this 2 days ago

sleimon added a commit that references this issue 9 hours ago
Added a new feature, lifetime wins/losses and session wins/losses 6b132e0

sleimon mentioned this 9 hours ago
[Added a new feature, lifetime wins/losses and session wins/losses #68](#)

sleimon 9 hours ago Owner

Game class is no longer overly dependent on the user class
Now the game class also relies on the player class as well

We solved this by splitting some of the user class functions and putting it in the player class so that the game class isn't overly dependent on the user class.

8) The game class has excessive access to the user class and database logic

Code Smell: Inappropriate Intimacy (Excessive Coupling Between Game and User Classes) #36

Closed

aryagith opened last week

Code Smell ID: 8

- **User Story:** User Profile
- **Description:**
`Game.java` has excessive access to `User` and database logic, creating strong dependencies that make modifications difficult. Additionally, `Main.java` improperly handles database operations.
- **Refactoring Suggestion:**
 - Use **Dependency Injection** to decouple `Game` from `User`.
 - Move **database logic** into a dedicated class and interact through **abstractions**.
- **Severity:** 🚨 High

Create sub-issue

aryagith added `code smell` last week

sleimon self-assigned this 2 days ago

sleimon mentioned this 9 hours ago

[Added a new feature, lifetime wins/losses and session wins/losses #68](#)

sleimon 9 hours ago Owner

There is very little database code in the game class (a couple lines in the constructor and one method) and there is no database code in the user class which means that a separate db logic class is pointless.

Now that the player class has a separate identity, the game class is no longer excessively accessing the user class.

Assignees: sleimon

Labels: code smell

Projects: No projects

Milestone: No milestone

Relationships: None yet

Development: [Create a branch](#)

Notifications: You're receiving notifications for this issue

Participants: sleimon, aryagith

→ Transfer issue

🔒 Lock conversation

📌 Pin issue

There is very little database code in the game class (a couple lines in the constructor and one method) and there is no database code in the user class which means that a separate db logic class is pointless.

Now that the player class has a separate identity which has absorbed some of the functions of the user class, the game class is no longer excessively accessing the user class.

9) Too many comments that explained what a method/class was doing

Code Smell: Comments #37

Closed



sleimon opened last week · edited by sleimon

Edits ...

Code Smell ID: 9

User Story: Blackjack Gameplay Logic

Description:

The classes involved in managing the gameplay logic contain many comments that mainly answer what the method does rather than the standard of why it's doing what it is.

Refactoring Suggestion:

Go back and rewrite comments so that it explains why a method is doing what it's doing.

Severity: ⚠ Low

Create sub-issue ...



sleimon added code smell last week



sleimon self-assigned this last week



sleimon added a commit that references this issue 2 days ago

Fixed some code smells  d540aa2



sleimon mentioned this 2 days ago

[Fixed some code smells #57](#)



sleimon 2 days ago

Owner Author ...

Removed any comments that explained what a method did and tried to put comments that explained why something did what it did

Solved this by removing all comments that said what something was doing rather than why it's doing it

10) A method in the person class called printHand() was out of place in its class because it handled some GUI functions

Code Smell: Feature Envy (GUI Logic in Person Class) #38

Closed

aryagith opened last week · edited by sleimon

Code Smell ID: 10

- **User Story:** Different Cards (Deck, Hand, and Game Functionality)
- **Description:** The `printHand()` method in `Person` manipulates GUI elements like `JLabel` and `ImageIcon`, violating **Single Responsibility Principle (SRP)**.
- **Refactoring Suggestion:** Move GUI logic into a dedicated `HandRenderer` or `GameUI` class to ensure `Person` remains focused on game logic.
- **Severity:** ⚠️ Medium

Create sub-issue

aryagith added code smell last week

sleimon self-assigned this 2 days ago

sleimon added 2 commits that reference this issue 2 days ago

Fixed some ticket #38, code smell ID #10 4dd0fd9

Merge pull request #59 from sleimon/sleiITR3 Verified 891ea41

sleimon 2 days ago Owner

Moved the `printHand()` method form the person class to the game class where currently all our gui is handled.

Solved this by moving the `printHand()` method from the person class to the game class where currently all our GUI is handled.

11) Some dead code in the stub database class

Code Smell : Redundant Code: Unused Code in StubDatabase Class

 Closed

 **roopsie** opened last week · edited by sleimon Edits ▾ ⋮

Code Smell ID: 14
User Story: StubDatabase Handling
Description:
There are large segments of code commented out and left in the StubDatabase class, which may have been intended for future use or extraction. This unused code unnecessarily clutters the class.
Refactoring Suggestion:
Delete the commented-out code that is no longer needed. This will improve the readability and cleanliness of the code and reduce unnecessary bloat.
Severity: Low

Create sub-issue ▾ 

  **roopsie** changed the title ~~Code-Smell~~ Code Smell : Redundant Code: Unused Code in StubDatabase Class last week

  **roopsie** added `code smell` last week

  **sleimon** added 2 commits that reference this issue 2 days ago

Fixed some ticket [#43](#), code smell ID [#14](#) df2a9af
Merge pull request [#61](#) from sleimon/sleiITR3 2778ed5 Verified

  **sleimon** self-assigned this 2 days ago

 **sleimon** 2 days ago Owner ⋮

Deleted the dead code and comments in the class



Solved this by removing the dead code

12) There is an if statement that is being spammed across the game class

Code Smell - Repeated Code #67

Closed



sleimon opened 11 hours ago

Code Smell ID: 15
User Story: StubDatabase Handling
Description:
In the game class, there is an if statement relating to the database that is frequently repeated
Refactoring Suggestion:
Put that code into a method and call the method when needed
Severity: Low

Create sub-issue



sleimon added `code smell` 11 hours ago



sleimon self-assigned this 11 hours ago



sleimon 11 hours ago

Owner Author

I put the code into a method and just called that method when needed

Solved this by putting the code into a method and just calling that method when needed

1)

[illegible]

2)

Bug: All the winning methods don't increase chip count #40

Closed



sleimon opened last week · edited by sleimon

Edits · ...

Problem Report Number: 4

Reported By: Sleimon Naimi

Date Reported: March/24th/2025.

Program/Component Name:

Path: 23118blackjackTeam4/src/main/java/com.blackjack/Models/Game.java.

Release Number: Released on March/7th/2025 at 1:58 P.M.

Version/Build Identifier: Git Hash: [82430e0](#)

Configurations:

Hardware:

- Device: IBUYPOWER Trace Mesh Gaming Desktop
- CPU: Intel(R) Core(TM) i7-14700F 2.10 GHz
- RAM: 32GB

Software:

- OS: Windows 11.
- Database: PostgreSQL on Supabase.
- Runtime Environment: Java JDK 21.



Report Type: Coding

Can Reproduce: Yes.

Severity: High

Priority: High

Problem Summary: Winning in any of the 5 ways of blackjack doesn't result in an increase of chips for the player

Key Words: Bet, Win, Increase, Chips

Problem description and how to reproduce it:

In any playthrough, try to win in any of these 5 ways:

1. Winning a hand normally by getting 21, having a higher hand value than the dealer, or the dealer busting
2. Drawing a blackjack when the dealer doesn't have one, known as an instant blackjack
3. Hitting the double down button and winning
4. Predicting the dealer has 21 and correctly choosing insurance resulting in a win for you
5. A tie with the dealer

When any of these happen, you can observe that your chip count never goes up

Suggested Fix: Taking a look at the logic methods in game.java. Specifically `initialBlackjack()`, `dealerInitialBlackjack()`, `playerTurn()`, `checkPlayer21()`, `dealerTurn()`, `non21win()`, and `insurance()` methods

Status: Closed

Resolution: Resolved

Resolved By: sleimon

Create sub-issue



sleimon added [bug](#) last week

sleimon self-assigned this last week

sleimon added 2 commits that reference this issue 5 days ago

Closed problem report [#4](#), issue [#40](#)

Merge pull request [#45](#) from sleimon/slei1TR3

738d0ff

Verified 723faed

sleimon closed this as [completed](#) 5 days ago



sleimon 5 days ago

Owner

Author

...

Fixed the issues but still need to delete the user class and clean up smell in the game class



3)

Bug: When winning an insurance bet, the text trails off the screen #6

Closed



sleimon opened 11 hours ago

Problem Report Number: 6

Reported By: Sleimon Naimi

Date Reported: March/29th/2025.

Program/Component Name:
Path: 2311BlackjackTeam4/src/main/java/com.blackjack/Models/Game.java.

Release Number: Released on March/7th/2025 at 1:58 P.M.

Version/Build Identifier: Git Hash: [82438e8](#)

Configurations:

Hardware:

- Device: IBUYPOWER Trace Mesh Gaming Desktop
- CPU: Intel(R) Core(TM) i7-14700F 2.10 GHz
- RAM: 32GB

Software:

- OS: Windows 11.
- Database: PostgreSQL on Supabase.
- Runtime Environment: Java JDK 21.

Report Type: Coding

Can Reproduce: Yes.

Severity: Low

Priority: Low

Problem Summary: Text on the screen informing the player they won an insurance bet trails off the screen.

Key Words: Text, Win, Insurance

Problem description and how to reproduce it:
In any playthrough, try to win a hand by predicting the dealer has drawn a blackjack

1. Load up game
2. Bet any amount
3. Successfully predict the dealer has drawn a blackjack
4. Hit the insurance button

Suggested Fix: Shorten text or adjust text box

Status: Closed

Resolution: Resolved

Resolved By: sleimon

Create sub-issue



sleimon added bug 11 hours ago



sleimon self-assigned this 11 hours ago



sleimon 11 hours ago

Owner Author

Shortened text to appropriate length



sleimon closed this as completed 11 hours ago

4)

BUG: Getting to 0 chips and hitting restart doesn't reset your chips back to the default amount, looping the game over screen #65

Closed



sleimon opened 11 hours ago

Problem Report Number: 7

Reported By: Sleimon Naimi

Date Reported: March/29th/2025.

Program/Component Name:

Path: 2311BlackjackTeam4/src/main/java/com.blackjack/Models/Game.java.

Release Number: Released on March/7th/2025 at 1:58 P.M.

Version/Build Identifier: Git Hash: [82430e0](#)

Configurations:

Hardware:

- Device: IBUYPOWER Trace Mesh Gaming Desktop
- CPU: Intel(R) Core(TM) i7-14700F 2.10 GHz
- RAM: 32GB

Software:

- OS: Windows 11.
- Database: PostgreSQL on Supabase.
- Runtime Environment: Java JDK 21.

Report Type: Coding

Can Reproduce: Yes.

Severity: Low

Priority: Low

Problem Summary: Hitting 0 chips triggers the game over screen. Hitting restart should reset chips back to the default amount but it doesn't continuously setting the chips to 0 and looping the game over screen.

Key Words: Game Over, 0 Chips, Loop

Problem description and how to reproduce it:

In any playthrough:

1. Load up the game
2. Hit the bet all button
3. Keep hitting until you bust
4. Observe the game over screen
5. Hit the restart button and the game over screen will keep looping, chips will continue to be 0

Suggested Fix: Look at the restart method

Status: Closed

Resolution: Resolved

Resolved By: sleimon

Create sub-issue



sleimon added [bug](#) 11 hours ago

sleimon self-assigned this 11 hours ago



sleimon 11 hours ago

Owner

Author

...

Added a line to the restart method that sets the player's chips to 1000



Assignees

sleimon

Labels

bug

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

Create a branch

Notifications

You're receiving
subscribed to the

Participants



Transfer issue

Lock conversation

Pin issue

Delete issue